KSHEER SAGAR AGRAWAL

• Applied AI • AI Systems & Infra • Software Engineering

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EDUCATION

University of California San Diego (UCSD) — GPA: 4.0/4.0
M.S. Electrical & Computer Engineering (Machine Learning & Data Science) — 2025-2027

Indian Institute of Technology, Gandhinagar (IITGN) — CPI: 8.7/10.0
B.Tech. Electrical Engineering with Minors in Computer Science and Engineering [Transcipt] — 2020-2024

Delhi Public School, Mumbai — Percentage: 94.0%
Class XII, Central Board of Secondary Education [Transcipt] — 2019-2020

Delhi Public School, Mumbai — Percentage: 95.8%
Class X, Central Board of Secondary Education [Transcipt] — 2017-2018

EXPERIENCES

Amazon Web Services | Software Development Engineer | DynamoDB

Mentor: Rahul Verma | DAX | In-Memory Cache | Certificate — *July '24 - Sept'25*

- Reduced operational access to production DynamoDB cache clusters by **93%** from 7 to 0.5 per week by designing an operational tooling service that replaced manual SSH with secure SSM workflows.
- Improved DynamoDB cache release cycles **3x**, from 84 to 28 days as pipeline owner by implementing end-to-end CI/CD automation and eliminating manual deployment bottlenecks across all production environments.
- Maintained **99.97% uptime** for AWS DynamoDB caching service by resolving **200+ high-severity** production incidents as primary/secondary on-call engineer, minimizing customer impact through rapid incident response.
- **Mentored a 6-month intern** who delivered an automated load test validation system with run-time metrics evaluation and pipeline-integrated release approval. **Intern received return offer.**
- Led migration of HostManager monitoring service from **Java 8** to **21** and **Spring Boot 3.5**, refactoring Mockito static tests, and building native executables using GraalVM for faster startup and lower memory footprint.
- **Achieved gold status with 1291 points** on Amazon's internal Stack Overflow.

Amazon | Software Development Intern | Amazon Go

Mentor: Rohit Sharma | IAT | Just Walk Out Technology | Certificate — *May '23 - July'23*

- **Eliminated Server maintenance overhead** by migrating AmazonGo stores device monitoring service from a monolithic job to AWS Lambdas with EventBridge triggers.
- **Reduced Lambda run time from 45 to 30 seconds** by refactoring Java codebase through abstract base class inheritance, removing unused serialization classes, and increasing test coverage from 80 to 97%.

ACADEMIC RESEARCH PROJECTS

LIPIDS: Learning-based Illumination Planning for Photometric Stereo

Research Publication • CVIG Lab • Advisor: Prof. Shanmuganathan Raman • Paper • Poster — *January'24 - May'24*

- Proposed a novel Light Sampling Network (LSNet) for optimal illumination planning in photometric stereo, enabling minimal-light configurations with near state-of-the-art accuracy.
- Achieved performance on par with online reinforcement learning approaches while reducing inference time from **20** seconds to $\sim$**1** second through an offline learning framework.

Effects of Structured Pruning on Handling Uncertainty Estimates

Research Project • Advisor: Prof. Nipun Batra • Poster • Report • Code — *August'23 - April'24*

- Achieved $\approx$ **80%** parameter reduction with $\leq$ **2%** accuracy drop using pruning strategies on CIFAR-10/100.
- Extended Lottery Ticket Hypothesis to structured pruning, preserving accuracy and improving calibration ($\downarrow$ **ECE/UCE** $\leq$**5%**)

DeblurGAN: Motion Deblurring Using Deep Generative Adversarial Network

Research Project • Advisor: Prof. Shanmuganathan Raman • Poster • Code — *January'23 - May'23*

- Implemented a **PyTorch-based GAN model** for motion deblurring inspired by the *Deep Generative Filter for Motion Deblurring* paper, integrating multiple performance-enhancing optimizations.
- Utilized **LPIPS loss**, **Differential Augmentation**, and **Wasserstein GAN loss** to improve color consistency, gradient stability, and training convergence.

SELECTED PROJECTS

Academic Request Database System

Institute Building Project • Supervisor: Prof. Dilip Srinivas Sundaram • Code *August'21 - December'21*

- Built Institute's academic portal (Node.js) handling **1K+** monthly requests (transcripts, fees, certificates).
- Reduced avg. turnaround to $\leq$ **2 days** through structured data collection (EJS), automated admin routing (AppScript) & email-based ticketing system.
- Technology Used: NodeJS, Express, Google Authentication, Google Sheets, Apps Script, Heroku Cloud Services.

ARM SentryRover

Course Project • Advisor: Prof. Jhuma Saha • Code *January'23 - April'23*

- Programmed Obstacle Detection Algo in **Embedded C** to interface GPS, ESP8266 camera sensors for navigation.
- Developed a web-application to control and monitor an STM-based surveillance robot using NodeMCU communication protocol and **STM32 (ARM Cortex-M)** microcontroller.

Statistical Language Modelling Using N-grams

Course Project • Advisor: Prof. Mayank Singh • Code *January'22 - March'22*

- Used the Trigram Model to predict the next word and find the perplexity of a given sentence.
- Technology Used : Python, Natural Language Toolkit, HuggingFace.

Image Segmentation using Random Walker Algorithm

Course Project • Advisor: Prof. Shanmuganathan Raman • Code *September'22 - October'22*

- Implemented the Random Walker Algorithm for image segmentation, involving marked and unmarked pixel nodes.
- Calculated the probability of neighboring pixels adopting the same value based on pixel intensity differences.
- Utilized the combinatorial Dirichlet problem solution to determine final probabilities for pixel labels.
- Performed matrix calculations, and constructed a new image from calculated probabilities with a pixel accuracy of 93.7%.

Real-Time Object Tracking using Gaussian Mixture Models

Course Project • Advisor: Prof. Shanmuganathan Raman • Code *October'22 - November'22*

- Processed and analyzed a 30-second traffic video from YouTube, using OpenCV for frame extraction.
- Applied GMM to model pixel values, enhancing background-foreground distinction for precise object tracking.

TEACHING EXPERIENCE

- **Certification in Teaching Program, IIT GN -** Completed a 6-week long program engaging in practice modules and seminars aimed at developing and enhancing teaching skills.
- **Teaching Assistant, CS432 Databases, IIT GN** *Jan'24 - May'24*
- **Teaching Assistant, ES242 Data Structures & Algorithms, IIT GN** *Aug'22 - Dec'22*
- **Teaching Assistant, Data Structures & Algorithms in C++, Coding Ninjas Ed.Tech Company:** *July'22 - Aug'22*
  Actively participated in resolving programming and debugging doubts for 100+ students at Coding Ninjas.

ACHIEVEMENTS AND LEADERSHIP

- **ICPC Regional Contestant** @ UC San Diego – Representing UCSD at the ICPC SoCal Regionals 2025.
- Achieved Top 10 selection in India for the prestigious **SIA Youth Scholarship '19** for pre-university studies in Singapore.
- Developed and deployed an institute-wide **Academic Request System**, used daily by 2000+ IITGN students.
- Received the **Dean's List Award** @ IITGN for outstanding academic performance in all eligible semesters.
- **Class Representative** @ Electrical Engineering IITGN – Coordinated with faculty to ensure smooth academic operations.
- **Problem Setter & Member** @ Competitive Programming Club – Conducted biweekly contests for the IITGN community.
- **Badminton Captain** @ IITGN – Led a 5-member team at the prestigious Inter-IIT Sports Meet'22.
- **Completed the New Delhi Marathon** (42.2 km) in 5 h 28 m without prior training.

TECHNICAL SKILLS

Languages: C++ (STL) | Python | Java | Ruby | SQL | C | Bash | CUDA | Verilog | SystemVerilog

Tools and Frameworks: Linux | Git | MongoDB | Django | Docker | Kubernetes | Jupyter | NumPy | Pandas | PyTorch | TensorFlow | JAX | Hugging Face | LLVM

Concepts: Distributed Systems | Parallel Computing | Event-Driven Architecture | Performance Profiling | Scalability | Runtime System | System Design | CI/CD Pipeline | Microservices | Compiler Optimization | Network | Database